public test blockchain. You'll have to repeat this operation to get more ether.

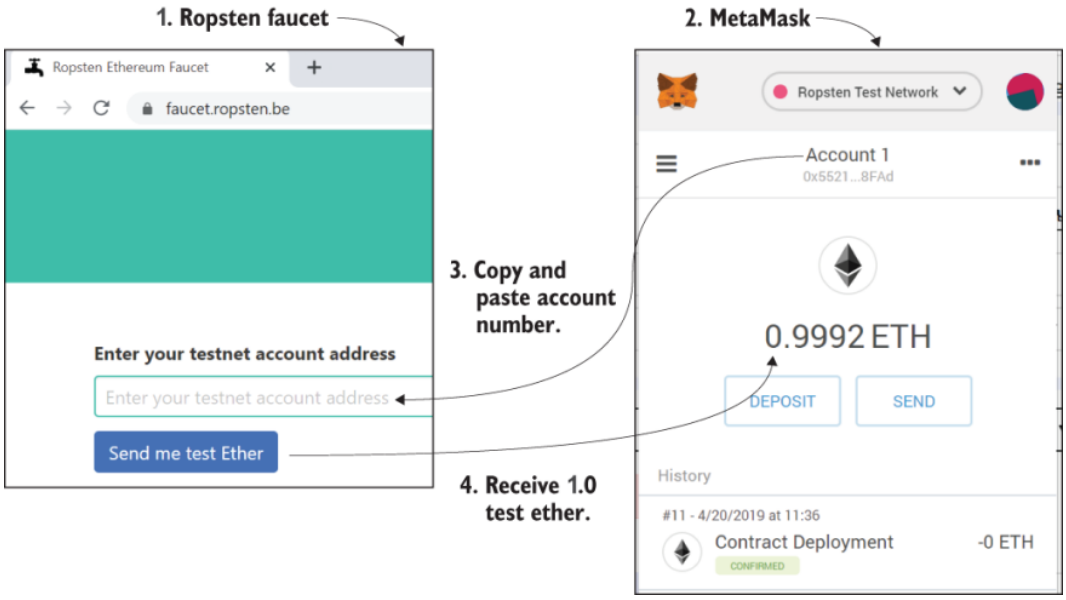



**Figure 5.6 Obtaining test ether from a Ropsten faucet**

## 5.2.6 Deploying and transacting on Ropsten

Let's use the simple Counter smart contract (listing 5.1) to demonstrate the deployment on Ropsten. By now, you must have realized that you need account balances in ether to deploy and operate on public blockchain networks. Every transaction costs ether, though typically only a fraction.

**Listing 5.1 Counter.sol**

```
pragma solidity >=0.4.21 <=0.6.0;
// Imagine a big integer counter that the whole world could share
contract Counter {
    uint value;

    function initialize (uint x) public {
        value = x;
    }

    function get() view public returns (uint) {
        return value;
    }

    function increment (uint n) public {
```